Divyesh Khandhedia

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SUMMARY

Technically proficient with experience of 6+ years in Power System Consulting, including planning/feasibility studies, load flow, short circuit, voltage stability, Transient/dynamic stability, Reactive power compensation, Capacitor/ Series, Shunt Reactor/ STATCOM sizing, PQ/PV/QV curve preparation, Grid code compliance/Grid integration studies for onshore and offshore plants, EMT - Electromagnetic Transients studies, Insulation coordination studies, TOV, and Power quality studies including harmonic analysis. Report writing to convey study results to client.

EXPERIENCE

GE Vernova, ***Application Engineer***
Aug 2024 - Present
• Renewable plants planning studies to assess adequacy. Grid Forming and Grid following controllers. Inverters and PPC Generic and UDM modeling.
• Advanced studies performed – SSTI, Small Signal Stability study.
• Grid Code compliance test for Power plants in different grid codes such as ERCOT. Reactive Power Compensation studies, Short Circuit studies, Transient Stability Studies.
• PSSE to PSCAD model conversion using ETRAN, PRSIM software.
• Industrial plant studies such as Transient stability, Islanding study, Critical Clearing Time (CCT) determination.
• TOV Temporary Overvoltage, TRV Transient Recovery Voltage Studies in PSCAD.
• Model Validation of Synchronous generators.
• Data Center load modeling using CMLD (Composite Load Modeling). Did in-depth understanding of load Modeling using Complex, Composite modeling.
• Harmonic Modeling and analysis.
• Motor Starting study and analysis in software's such as PSSE, PSLF, ETAP and PSCAD. Performed comparison to benchmark results across different software environment.
• Governor FGMO testing and model validation and Synchronous machine model validation for Excitation model.

Burns & McDonnell, ***Associate Senior Electrical Engineer***
Jan 2021 - Aug 2024
• Perform steady state and dynamic modeling of different types of generation systems such as Solar, Wind, BESS, Synchronous. Tuning of controls wherever required to meet requirements as per guidelines.
• Relationship with different Inverter manufacturers such as Sungrow, TMEIC, Power Electronics, Tesla, Dynapower, GE Vernova, Siemens etc. and PPC manufacturers such as Norcal, Power Electronics, Sungrow, Siemens etc. for queries resolutions.
• Offshore wind generation planning and transient stability studies. HVDC and HVAC power evacuation strategies.
• Sizing of compensation with STATCOMs, shunt reactor and capacitor banks for various plants visited database columns, which resulted in a 25% reduction in response times.
• Short Circuit mitigation Strategies using Series reactors, IS Limiter and its selection.
• Leading a team of 5 people. I fast adopted to new projects and took a lead. Got promoted from Engineer to Associate Engineer in a year.

Suvidha Electricals, ***Engineer***
Aug 2020 - Jan 2021
• Industrial Power system studies, Motor Starters like DOL, Star Delta and Soft starter, VFD (Variable Frequency Drive) Wiring, Process of Fabrication of panels, Assembly of panels, Power & Control Wirings.
• Backup power auto-changeover methodology. Troubleshooting in industrial sites.

Larsen & Toubro, ***Apprentice***
Dec 2016 - Jun 2017
• Design and development of MCCB, Spring tension evaluation, Prototyping and validation of Contact pressure,
• Testing of Circuit Breaker as per IEC 60947-1&2.
• Brief knowledge about Contactors, Relays, Air circuit breakers, MCB, Busbars, RCCB, and Electric Panels.

Reliance Energy, ***Apprentice***
Jun 2015 - Dec 2015
• Power Generation using Thermal energy, Distribution system reliability,

• Operation and Maintenance of Distribution systems. Fault Detection and Repairing Methods, Power distribution Infrastructure, Process of fault repairing, Load shifting and Maintenance of (HV & MV transformers) etc.

EDUCATION

University of Mumbai, MH, India
Bachelors in Electrical Engineering
• Graduated with First Class
• Obtained two internships in MAHATRANSCO (Maharashtra state transmission company), Central Railways, Western Railways during Summer and Winter Vacations.
• Courses: Power Systems, Protections, Electrical Machines etc.
• Participated in Smart India Hackathon(Government organized event) in the year 2018 event runner up.
• Again, participated in Next year Hackathon event and Won Smart India Hackathon competition in the year 2019.

Shri Bhagubhai Mafatlal Polytechnic, MH, India
Diploma in Electrical Engineering
• Topper of the class, Did two Apprenticeships in Reliance and Larsen & Turbro during this tenure as per course requirement for duration of 6 months.
• Courses: Basics of Power Systems, Protections and Control Systems, Electrical Machines etc.

LICENSES & CERTIFICATIONS

Electric Utilities Fundamentals and Future
Coursera (University of Colorado) •

Electric Power Systems
Coursera (University at Buffalo and The State University of New York) •

Solar Energy and Electrical System Design
Coursera (University at Buffalo) •

SKILLS

PSSE • PSLF • PSCAD • Digsilent Powerfactory • ETAP • Aspen Oneliner • Powergem tara • English • Hindi

HONORS & AWARDS

impact award
for completing the projects within a strict timeline